

Loading Silver Layer

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Stored Procedure: Load Silver Layer (Bronze -> Silver)
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Script Purpose:
    This stored procedure performs the ETL process to populate the
    'silver' schema tables from the 'bronze' schema.

    Actions Performed:
        - Truncates Silver tables.
        - Inserts transformed and cleansed data from Bronze into Silver tables.
        - Calculates load duration for each table.
        - Calculates total Silver Layer load duration.
        - Handles errors using TRY...CATCH.

Parameters:
    None.

Usage Example:
    EXEC silver.load_silver;
=====
*/

CREATE OR ALTER PROCEDURE silver.load_silver
AS
BEGIN

    DECLARE @start_time DATETIME;
    DECLARE @end_time DATETIME;
    DECLARE @batch_start_time DATETIME;
    DECLARE @batch_end_time DATETIME;

    BEGIN TRY

        SET @batch_start_time = GETDATE();

        PRINT '=====';
        PRINT 'Loading Silver Layer';
        PRINT '=====';

        -----
        -- Loading silver.crm_cust_info
        -----

        PRINT '=====';
        PRINT 'Loading Table: silver.crm_cust_info';
        PRINT '=====';

        SET @start_time = GETDATE();

        PRINT '>> Truncating Table: silver.crm_cust_info';

        TRUNCATE TABLE silver.crm_cust_info;

        PRINT '>> Inserting Data Into: silver.crm_cust_info';

        INSERT INTO silver.crm_cust_info (
            cst_id,
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        cst_key,
        cst_firstname,
        cst_lastname,
        cst_marital_status,
        cst_gndr,
        cst_create_date
    )
SELECT
    cst_id,
    cst_key,
    TRIM(cst_firstname) AS cst_firstname,
    TRIM(cst_lastname) AS cst_lastname,

    CASE
        WHEN UPPER(TRIM(cst_marital_status)) = 'S' THEN 'Single'
        WHEN UPPER(TRIM(cst_marital_status)) = 'M' THEN 'Married'
        ELSE 'n/a'
    END AS cst_marital_status,

    CASE
        WHEN UPPER(TRIM(cst_gndr)) = 'F' THEN 'Female'
        WHEN UPPER(TRIM(cst_gndr)) = 'M' THEN 'Male'
        ELSE 'n/a'
    END AS cst_gndr,

    cst_create_date

FROM (
    SELECT
        *,
        ROW_NUMBER() OVER (
            PARTITION BY cst_id
            ORDER BY cst_create_date DESC
        ) AS flag_last

    FROM bronze.crm_cust_info

    WHERE cst_id IS NOT NULL
) t

WHERE flag_last = 1;

SET @end_time = GETDATE();

PRINT '>> Load Duration: '
+ CAST(DATEDIFF(MILLISECOND, @start_time, @end_time) AS NVARCHAR)
+ ' milliseconds';

PRINT '>> -----';

-----
-- Loading silver.crm_prd_info
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PRINT '=====';
PRINT 'Loading Table: silver.crm_prd_info';
PRINT '=====';

SET @start_time = GETDATE();

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PRINT '>> Truncating Table: silver.crm_prd_info';

TRUNCATE TABLE silver.crm_prd_info;

PRINT '>> Inserting Data Into: silver.crm_prd_info';

INSERT INTO silver.crm_prd_info (
    prd_id,
    cat_id,
    prd_key,
    prd_nm,
    prd_cost,
    prd_line,
    prd_start_dt,
    prd_end_dt
)
SELECT
    prd_id,

    -- Extract category ID
    REPLACE(
        SUBSTRING(prd_key, 1, 5),
        '-', '_'
    ) AS cat_id,

    -- Extract product key
    SUBSTRING(
        prd_key,
        7,
        LEN(prd_key)
    ) AS prd_key,

    prd_nm,

    -- Replace NULL product cost with 0
    ISNULL(prd_cost, 0) AS prd_cost,

    -- Map product line codes to descriptive values
    CASE
        WHEN UPPER(TRIM(prd_line)) = 'M' THEN 'Mountain'
        WHEN UPPER(TRIM(prd_line)) = 'R' THEN 'Road'
        WHEN UPPER(TRIM(prd_line)) = 'S' THEN 'Other Sales'
        WHEN UPPER(TRIM(prd_line)) = 'T' THEN 'Touring'
        ELSE 'n/a'
    END AS prd_line,

    CAST(prd_start_dt AS DATE) AS prd_start_dt,

    /*
        Calculate the product end date:
        - LEAD() gets the next product version's start date.
        - PARTITION BY prd_key keeps each product's history separate.
        - ORDER BY prd_start_dt sorts versions by start date.
        - Subtract 1 day so the current version ends
          one day before the next version starts.
        - CAST(... AS DATE) converts the result to DATE.
    */
    CAST(
        LEAD(prd_start_dt) OVER (
            PARTITION BY prd_key

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        ORDER BY prd_start_dt
    ) - 1 AS DATE
) AS prd_end_dt

FROM bronze.crm_prd_info;

SET @end_time = GETDATE();

PRINT '>> Load Duration: '
+ CAST(DATEDIFF(MILLISECOND, @start_time, @end_time) AS NVARCHAR)
+ ' milliseconds';

PRINT '>> -----';

-----
-- Loading silver.crm_sales_details
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PRINT '=====';
PRINT 'Loading Table: silver.crm_sales_details';
PRINT '=====';

SET @start_time = GETDATE();

PRINT '>> Truncating Table: silver.crm_sales_details';

TRUNCATE TABLE silver.crm_sales_details;

PRINT '>> Inserting Data Into: silver.crm_sales_details';

INSERT INTO silver.crm_sales_details (
    sls_ord_num,
    sls_prd_key,
    sls_cust_id,
    sls_order_dt,
    sls_ship_dt,
    sls_due_dt,
    sls_sales,
    sls_quantity,
    sls_price
)
SELECT
    sls_ord_num,
    sls_prd_key,
    sls_cust_id,

    CASE
        WHEN sls_order_dt = 0 OR LEN(sls_order_dt) != 8
            THEN NULL
        ELSE CAST(CAST(sls_order_dt AS VARCHAR) AS DATE)
    END AS sls_order_dt,

    CASE
        WHEN sls_ship_dt = 0 OR LEN(sls_ship_dt) != 8
            THEN NULL
        ELSE CAST(CAST(sls_ship_dt AS VARCHAR) AS DATE)
    END AS sls_ship_dt,

    CASE

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        WHEN sls_due_dt = 0 OR LEN(sls_due_dt) != 8
            THEN NULL
        ELSE CAST(CAST(sls_due_dt AS VARCHAR) AS DATE)
    END AS sls_due_dt,

    CASE
        WHEN sls_sales IS NULL
            OR sls_sales <= 0
            OR sls_sales != sls_quantity * ABS(sls_price)
            THEN sls_quantity * ABS(sls_price)
        ELSE sls_sales
    END AS sls_sales,

    sls_quantity,

    CASE
        WHEN sls_price IS NULL OR sls_price <= 0
            THEN sls_sales / NULLIF(sls_quantity, 0)
        ELSE sls_price
    END AS sls_price

FROM bronze.crm_sales_details;

SET @end_time = GETDATE();

PRINT '>> Load Duration: '
+ CAST(DATEDIFF(MILLISECOND, @start_time, @end_time) AS NVARCHAR)
+ ' milliseconds';

PRINT '>> -----';

-----
-- Loading silver.erp_cust_az12
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PRINT '=====';
PRINT 'Loading Table: silver.erp_cust_az12';
PRINT '=====';

SET @start_time = GETDATE();

PRINT '>> Truncating Table: silver.erp_cust_az12';

TRUNCATE TABLE silver.erp_cust_az12;

PRINT '>> Inserting Data Into: silver.erp_cust_az12';

INSERT INTO silver.erp_cust_az12 (
    cid,
    bdate,
    gen
)
SELECT
    CASE
        WHEN cid LIKE 'NAS%'
            THEN SUBSTRING(cid, 4, LEN(cid))
        ELSE cid
    END AS cid,

    CASE

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        WHEN bdate > GETDATE()
        THEN NULL
    ELSE bdate
END AS bdate,

CASE
    WHEN UPPER(TRIM(gen)) IN ('F', 'FEMALE')
    THEN 'Female'
    WHEN UPPER(TRIM(gen)) IN ('M', 'MALE')
    THEN 'Male'
    ELSE 'n/a'
END AS gen

FROM bronze.erp_cust_az12;

SET @end_time = GETDATE();

PRINT '>> Load Duration: '
+ CAST(DATEDIFF(MILLISECOND, @start_time, @end_time) AS NVARCHAR)
+ ' milliseconds';

PRINT '>> -----';

-----
-- Loading silver.erp_loc_a101
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PRINT '=====';
PRINT 'Loading Table: silver.erp_loc_a101';
PRINT '=====';

SET @start_time = GETDATE();

PRINT '>> Truncating Table: silver.erp_loc_a101';

TRUNCATE TABLE silver.erp_loc_a101;

PRINT '>> Inserting Data Into: silver.erp_loc_a101';

INSERT INTO silver.erp_loc_a101 (
    cid,
    cntry
)
SELECT
    REPLACE(cid, '-', '') AS cid,

    CASE
        WHEN TRIM(cntry) = 'DE'
        THEN 'Germany'

        WHEN TRIM(cntry) IN ('US', 'USA')
        THEN 'United States'

        WHEN TRIM(cntry) = '' OR cntry IS NULL
        THEN 'n/a'

        ELSE TRIM(cntry)
    END AS cntry
END AS cntry

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FROM bronze.erp_loc_a101;

SET @end_time = GETDATE();

PRINT '>> Load Duration: '
      + CAST(DATEDIFF(MILLISECOND, @start_time, @end_time) AS NVARCHAR)
      + ' milliseconds';

PRINT '>> -----';

-----
-- Loading silver.erp_px_cat_glv2
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PRINT '=====';
PRINT 'Loading Table: silver.erp_px_cat_glv2';
PRINT '=====';

SET @start_time = GETDATE();

PRINT '>> Truncating Table: silver.erp_px_cat_glv2';

TRUNCATE TABLE silver.erp_px_cat_glv2;

PRINT '>> Inserting Data Into: silver.erp_px_cat_glv2';

INSERT INTO silver.erp_px_cat_glv2 (
    id,
    cat,
    subcat,
    maintenance
)
SELECT
    id,
    cat,
    subcat,
    maintenance

FROM bronze.erp_px_cat_glv2;

SET @end_time = GETDATE();

PRINT '>> Load Duration: '
      + CAST(DATEDIFF(MILLISECOND, @start_time, @end_time) AS NVARCHAR)
      + ' milliseconds';

PRINT '>> -----';

-----
-- Completion
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SET @batch_end_time = GETDATE();

PRINT '=====';
PRINT 'Silver Layer Loading Completed';
PRINT '=====';

PRINT '>> Total Load Duration: '

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        + CAST(
            DATEDIFF(
                MILLISECOND,
                @batch_start_time,
                @batch_end_time
            ) AS NVARCHAR
        )
        + ' milliseconds';

PRINT '=====';

END TRY

BEGIN CATCH

PRINT '=====';
PRINT 'ERROR OCCURRED DURING SILVER LAYER LOADING';
PRINT '=====';

PRINT 'Error Message: ' + ERROR_MESSAGE();
PRINT 'Error Number: ' + CAST(ERROR_NUMBER() AS NVARCHAR);
PRINT 'Error State: ' + CAST(ERROR_STATE() AS NVARCHAR);
PRINT 'Error Line: ' + CAST(ERROR_LINE() AS NVARCHAR);

PRINT '=====';

END CATCH

END;
GO

-----
-- Execute Stored Procedure
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EXEC silver.load_silver;

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